

IILM UNIVERSITY

GREATER NOIDA

INTERNSHIP / JOB SIMULATION REPORT

Advanced Software Engineering Job Simulation



Roll No.: 25SCS1003003690

Section: 1CSE29

Programme: B.Tech – Computer Science and Engineering



Certificate / Internship Details

This report is prepared from the Certificate of Completion for the Advanced Software Engineering Job Simulation.

Student Name: Krish Raj

Roll No.: 25SCS1003003690

Section: 1CSE29

University: IILM UNIVERSITY, GREATER NOIDA

Simulation: Advanced Software Engineering Job Simulation

Period: September 2026

Issuing Organization: Forage

Student Declaration



I, Krish Raj, Roll No. 25SCS1003003690, Section 1CSE29, student of IILM UNIVERSITY, GREATER NOIDA, declare that this report presents the work and learning activities associated with the Advanced Software Engineering Job Simulation completed during September 2026.

The report has been prepared for academic documentation and reflects the practical task areas stated on the Certificate of Completion.

Student Signature: _____

Name: Krish Raj

Roll No.: 25SCS1003003690

Section: 1CSE29



Acknowledgement

I would like to express my sincere gratitude to IILM UNIVERSITY, GREATER NOIDA for providing an academic environment that encourages practical learning and professional development.

I am also thankful to Forage for providing the Advanced Software Engineering Job Simulation, which gave me an opportunity to work through practical task areas related to software engineering and computer science.

The simulation helped me connect academic concepts with practical problem-solving and gave me a better understanding of the importance of structured thinking in software development.

Krish Raj | Roll No.: 25SCS1003003690 | Section: 1CSE29



1. Introduction & 2. Objectives

The Advanced Software Engineering Job Simulation was completed by Krish Raj during September 2026.

The simulation involved practical tasks in four major areas: Advanced Data Structures, Software Architecture, Relational Database Design, and Data Munging.

Purpose: document the simulation, summarize the four task areas, and describe the learning outcomes.

Objectives: gain practical exposure; strengthen problem-solving and analytical thinking; understand advanced data structures; understand software architecture; learn relational database design; understand data preparation and transformation.



3. Advanced Data Structures & 4. Software Architecture

Advanced Data Structures: organize, store, and process data efficiently.

- Data organization and efficient access.
- Selection of suitable structures for different problems.
- Logical and systematic problem solving.

Software Architecture: high-level structure of a software system and how major components communicate.

- Breaking a system into understandable components.
- Defining responsibilities of major components.
- Designing software in a structured and maintainable way.



5. Relational Database Design & 6. Data Munging

Relational Database Design focuses on organizing information into related tables and defining how data items are connected.

- Organizing data into structured tables.
- Identifying relationships between data.
- Reducing unnecessary duplication of information.

Data Munging refers to preparing and transforming data into a useful form.

- Preparing data for further use.
- Transforming information into a consistent structure.
- Handling data in a systematic manner.



7. Skills and Learning Outcomes

Improved understanding of practical software-engineering workflows.

Better structured problem-solving approach.

Greater awareness of efficient data organization.

Basic understanding of designing software at a system level.

Improved understanding of structured relational data.

Exposure to practical data preparation and transformation.

Improved confidence in connecting academic concepts with practical tasks.



8. Conclusion

The Advanced Software Engineering Job Simulation provided practical exposure to four important areas: Advanced Data Structures, Software Architecture, Relational Database Design, and Data Munging.

Completing these task areas helped build a broader understanding of how different parts of software engineering work together.

The simulation strengthened technical thinking, practical software understanding, and professional learning documentation.

The Certificate of Completion confirms successful completion of the practical tasks during September 2026.



9. References / Supporting Document

Certificate of Completion – Advanced Software Engineering Job Simulation, issued by Forage, September 10, 2026.

Appendix: Certificate of Completion.